

Stage 2 — The location law and its mechanism

Where the Shortcut Sits: supervisor meeting 2 of 6

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In one sentence. When only the artifact’s overlap with the region of interest (ROI) is varied, ROI masking helps while the artifact lies outside the ROI and stops helping — in most sweeps it hurts — once the artifact overlaps it; the reversal comes from the artifact *surviving* the mask and dominating what the model sees, not from occlusion, and it recurs in ultrasound, capsule endoscopy and chest radiography with one principled exception.

Notation

$Y \in \{0, 1\}$	diagnosis (1 = melanoma / malignant / solid-component tumour / erosion).
$A \in \{0, 1\}$	artifact present (hair, sonographer caliper, capsule debris, chest tube).
ROI	region of interest (lesion, nodule, tumour, lesion box, lungs); drawn in green; artifacts in red.
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (in-ROI); Trap B: $r < 0.1$ (out-of-ROI).
Environments	training and correlated test $P(A=1 Y=1) = 0.9$, $P(A=1 Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real traps: half of each class; sweeps: absent).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ of reversed-test AUROC (real traps: Trap B – Trap A; sweeps and transplant: location interaction on the same images).
CI	95% crossed seed×image bootstrap interval (Stage 1, Sec. ??); every interval in these reports uses it.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted.
SMD	standardised mean difference between two groups.

1 Where we are

Stage 1 fixed the protocol (trap environments, frozen foundation-model features, linear heads, grouped splits, pre-registration, crossed bootstrap) and showed that the pilot’s central sweep result replicates. This meeting asks *why* masking reverses, and *how general* the reversal is. Everything in this report comes from runs regenerated in `results/rerun_2026-09-28/` and every interval uses the corrected (crossed seed × image) bootstrap. Relative to earlier drafts, three things are new: (i) the mechanism analyses now carry corrected intervals, and two lesion-size cells lost significance (Sec. 6); (ii) the same-head counterfactual is tested with a confidence interval instead of being shown as a figure only; (iii) the controlled sweep is repeated in four further modalities with two backbones

each where possible, and the counts of sweeps that support each part of the law are generated from the result files.

2 Question

2.1 What we want to know

- Q2a** Is the benefit of masking a function of the artifact’s overlap r with the ROI, with a sign change between $r = 0$ and $r = 1$? We call a positive *location interaction* $[\text{mask} - \text{ERM}]_{r=0} - [\text{mask} - \text{ERM}]_{r=1}$ on the reversed test the *location law*.
- Q2b** Which mechanism produces the reversal? Three candidates were stated in advance: *occlusion* (masking removes useful context whatever the artifact does), *lost disease signal* (the background carries diagnostic information), and *retention/concentration* (the part of the artifact inside the ROI survives masking and becomes a larger share of the visible pixels, so the head relies on it more).
- Q2c** Does the law hold beyond one backbone, one artifact rendering and one modality — and where does it fail?

2.2 Why these experiments, and what we rejected

- *Occlusion control instead of arguing from intuition.* If masking harmed because it throws away context, it would harm even when the artifact carries no label information (cf. Fong and Vedaldi, 2019). The control keeps the ruler at the same positions but makes it independent of the label.
- *Dilation as a dose of retention.* Growing the mask by a margin keeps more of the ruler visible without changing the lesion. If retention drives the harm, harm should grow with retention when retention is incomplete (50% overlap) and stay flat when the ruler is already fully retained (100% overlap).
- *Same-head counterfactual instead of saliency maps.* Saliency maps are qualitative and unstable for frozen encoders. We instead insert and remove the artifact on the *same* test image and read the change in the predicted probability $|\Delta p|$ of each trained head: a direct, per-image measure of reliance on the artifact.
- *Lesion size as a natural dose.* The same ruler covers a larger share of a small lesion; if concentration matters, harm should be largest for small lesions. A fixed ratio threshold was *not* pre-specified, because the pilot showed none.
- *Rejected:* measuring reliance through head weights (not comparable between masked and unmasked feature spaces) and a single “average” overlap (it would hide the sign change).

3 Methods

Controlled sweep ISIC 2018 pilot cohort (Stage 1), a 65×19 px ruler pasted at a frozen position per image and overlap $r \in \{0, 0.25, 0.5, 0.75, 1\}$ (Figure 2); DINOv2 ViT-B/14 at 518 px (Oquab et al., 2024), CLS features, logistic heads; seeds 42, 123, 456; environments 0.9/0.1 (training, correlated test), 0.1/0.9 (reversed test) and artifact-free (clean test). The primary outcome is the reversed-test AUROC.

Arms ERM (unmasked image); **ROI masking** (pixels outside the lesion mask set to the image mean before feature extraction); oracle inpainting (the known ruler pixels are replaced by the surrounding skin — an upper bound that needs the artifact mask); group-balanced reweighting of the artifact $\times$ label cells; deep feature reweighting (DFR) on a balanced subset (Kirichenko et al.,

2023); paired LEACE (Belrose et al., 2023) (the linear concept direction of the artifact is erased using images with and without it); GroupDRO (Sagawa et al., 2020). All arms share features and seeds; only the input or the head changes.

Occlusion control The ruler is pasted at the same positions in 50% of images independently of the label; AUROC is measured on a 50/50 test.

Mask dilation The lesion mask is dilated by 0, 10, 25 or 50 px before masking. *Retention* is the share of ruler pixels that remain visible; *ruler share* is the ruler’s share of all visible pixels.

Same-head counterfactual For each artifact-bearing image of the reversed test and each trained head, the image is scored with and without the pasted ruler (the masked head sees the masked version of both); $|\Delta p|$ is the absolute change in predicted probability. The mask – ERM difference of mean $|\Delta p|$ is bootstrapped with seeds and images crossed.

Lesion-size strata Test images are split into tertiles of lesion area; the median ruler-to-lesion area ratio is reported per tertile.

Replications and other modalities DINOv2 at 224 px, the dermatology foundation model DermLIP at 224 px (Yan et al., 2025), and a ruler whose colour, tick pattern and size are randomised per image (“stress”). Other modalities, each with its own ROI masks: thyroid nodules (TN3K (Gong et al., 2021); synthetic caliper; DINOv2 and MedSigLIP-448), ovarian tumours (MMOTU (Zhao et al., 2022); synthetic caliper; DINOv2), capsule-endoscopy lesions (SEE-AI (Yokote et al., 2023); synthetic debris; box ROIs; DINOv2 and MedSigLIP) and pneumothorax radiographs (NIH (Wang et al., 2017); synthetic tube; lung masks; RAD-DINO (Pérez-García et al., 2025) and DINOv2).

Statistics Every interval is a 95% percentile interval from 10,000 crossed bootstrap replicates (seeds resampled, one Poisson(1) weight per test image shared across seeds and arms). A claim is *supported* if its pre-stated criterion holds with the interval; “significant” below means the 95% interval excludes zero, not a multiplicity-adjusted test (the confirmatory family is in Stage 3).

4 Experiments

Table 1 lists each experiment of this stage with its registration status, the first commit that contains the registration and the support criterion. The pilot analyses (E1–E7) were specified before this project’s re-implementation (docs/REPLICATION_SPEC.md, 2699d51, 2026-09-26 15:47) and regenerated under the final registration (c45d482, 2026-09-28 00:35). **Caveat:** commit times come from this machine and are not independent proof of order.

Table 1: Experiments of this stage: registration status, the commit that first contains the registration, and what was stated in advance to count as support. Commit times come from this machine and are not independent proof of order.

Experiment	Status	Registration	First commit (local time)	Support criterion
Controlled dermoscopy sweep (E1)	supported registered	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 interaction CI > 0; mask-ERM CI > 0 at 0% and < 0 at 100%
Replications: DI-NOv2@224, DermLIP, randomised ruler	supported registered	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 same signs and verdicts as E1
Occlusion control (E4)	supported registered	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 no harm on the 50/50 test at any overlap
Mask dilation (E5)	supported registered	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 harm grows with retention at 50%; flat at 100%
Same-head counterfactual $ \Delta p $ (E6, T3)	supported registered	REVIEW2 (T3)	29cd0da, 18:56	2026-09-27 mask > ERM at in-ROI overlap, CI > 0
Lesion-size tertiles (E7)	partly supported pilot analysis, replicated	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 harm largest in small lesions
Thyroid caliper sweep	supported registered	THYROID_TRAPS	ea57661, 22:46	2026-09-26 interaction CI > 0
Ovarian caliper sweep	supported registered	OVARY_TRAPS	2517600, 07:29	2026-09-27 interaction CI > 0
Capsule debris sweep	supported registered	CAPSULE_TRAPS	fcbf8d9, 00:40	2026-09-27 interaction CI > 0
Chest tube sweep (RAD-DINO)	supported registered	CXR_DEVICE_TRAPS	4d1a924, 18:07	2026-09-26 interaction CI > 0
Chest tube sweep (DI-NOv2)	not supported post hoc	—	—	descriptive only
Regeneration with the crossed bootstrap (A1)	supported registered	FINAL (A1)	c45d482, 00:35	2026-09-28 a claim whose new CI includes 0 is rewritten

5 Results

5.1 The controlled sweep: the location law

Masking changes the reversed-test AUROC by +0.184 [+0.117, +0.250] when the ruler lies outside the lesion, by −0.097 [−0.164, −0.030] at 50% overlap and by −0.129 [−0.195, −0.064] at full overlap (Table 2, Figure 1); the location interaction is +0.313 [+0.258, +0.370]. Masking therefore lowers it significantly at 50, 75 and 100% overlap. Every other arm helps at every overlap, and their benefit *grows* with overlap — the opposite of masking — because they target the artifact, not the background. On the correlated test the masked model at full overlap scores 0.961 against 0.937 for ERM: the masked model exploits the ruler more, which is what the reversed test then punishes.

Table 2: Controlled dermoscopy sweep (ISIC 2018, DINOv2 ViT-B/14 @518, three seeds): change in reversed-test AUROC versus ERM, crossed CIs.

	0% overlap	50%	100%
ERM (reversed AUROC)	0.665	0.573	0.515
ROI masking – ERM	+0.184 [+0.117, +0.250]	−0.097 [−0.164, −0.030]	−0.129 [−0.195, −0.064]
oracle inpainting – ERM	+0.141 [+0.109, +0.175]	+0.211 [+0.175, +0.249]	+0.243 [+0.205, +0.284]
group-balanced – ERM	+0.137 [+0.099, +0.179]	+0.219 [+0.177, +0.265]	+0.281 [+0.236, +0.329]
DFR – ERM	+0.124 [+0.075, +0.179]	+0.221 [+0.166, +0.281]	+0.279 [+0.220, +0.339]
paired LEACE – ERM	+0.167 [+0.126, +0.211]	+0.237 [+0.197, +0.280]	+0.294 [+0.249, +0.339]
GroupDRO – ERM	+0.080 [+0.044, +0.118]	+0.152 [+0.112, +0.193]	+0.197 [+0.155, +0.240]

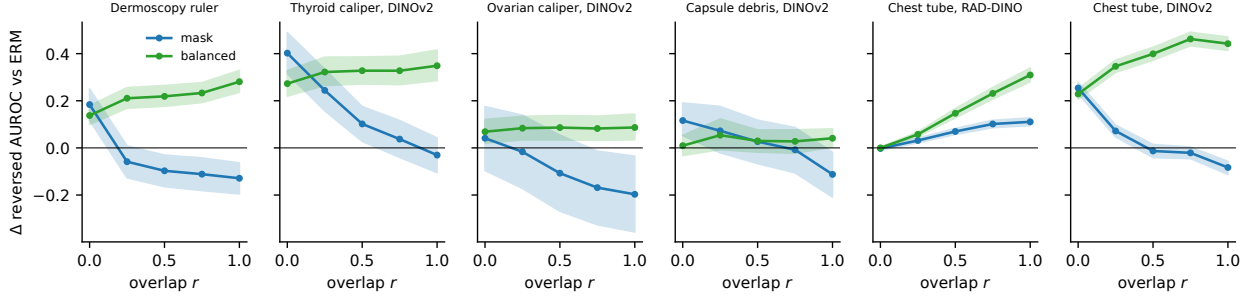


Figure 1: Change in reversed-test AUROC against ERM as a function of overlap for masking and group balancing (shaded: crossed 95% CI). Dermoscopy and the DINOv2/RAD-DINO sweeps in the other modalities.

Replications The sweep reverses with DINOv2 at 224 px (interaction $+0.456 [+0.390, +0.525]$), with DermLIP ($+0.424 [+0.335, +0.517]$) and with a randomised ruler ($+0.302 [+0.248, +0.357]$) (Table 3). The law is therefore not a property of one encoder, one resolution or one rendering of the ruler.

Table 3: Replications of the controlled sweep (reversed-test AUROC, crossed CIs).

Backbone / artifact	mask – ERM at 0%	at 100%	Location interaction
DINOv2 @224	$+0.361 [+0.275, +0.444]$	$-0.095 [-0.156, -0.035]$	$+0.456 [+0.390, +0.525]$
DermLIP @224	$+0.161 [+0.102, +0.222]$	$-0.263 [-0.348, -0.180]$	$+0.424 [+0.335, +0.517]$
DINOv2 @518, randomised ruler	$+0.146 [+0.082, +0.211]$	$-0.155 [-0.228, -0.084]$	$+0.302 [+0.248, +0.357]$

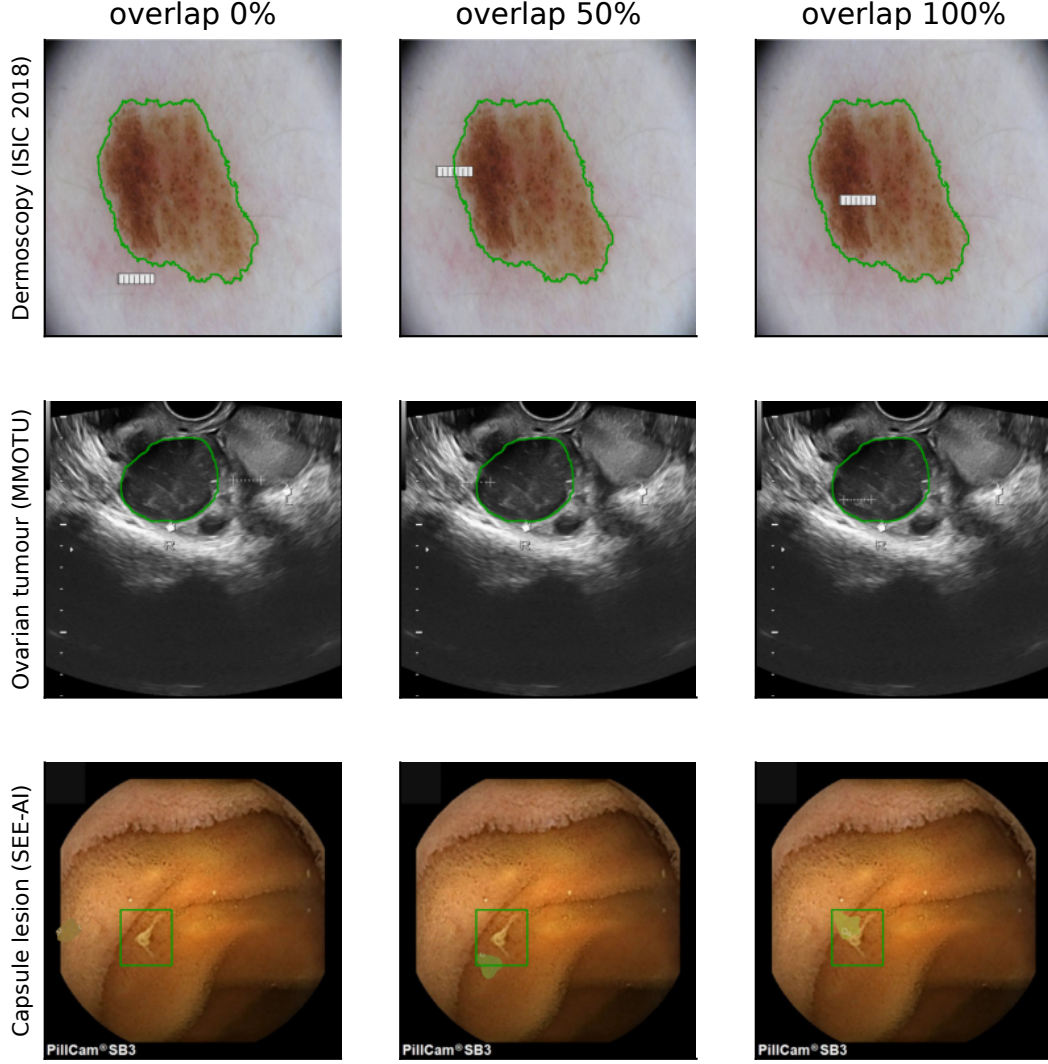


Figure 2: Real images with the synthetic artifact at 0, 50 and 100% overlap (green: ROI; the capsule ROI is the annotated bounding box). Top: ISIC 2018 (CC BY-NC 4.0); middle: MMOTU ovarian ultrasound (CC BY 4.0); bottom: SEE-AI capsule endoscopy (CC BY 4.0). Thyroid (TN3K) and NIH images are not shown because their redistribution terms are unclear.

5.2 Mechanism

Occlusion does not explain the harm With the ruler present but uninformative, masking changes it significantly at none of the five overlaps (every CI includes zero) (Table 4); inpainting, which removes only the ruler, changes nothing either. Masking by itself does not cost the reversed-test AUROC it loses in the sweep.

Table 4: Occlusion control: ruler present but uncorrelated with the label (50/50); AUROC change on the 50/50 test.

Overlap	mask – ERM	inpaint – ERM
0%	+0.036 [−0.016, +0.086]	−0.001 [−0.005, +0.003]
25%	+0.022 [−0.030, +0.073]	−0.010 [−0.034, +0.002]
50%	+0.041 [−0.010, +0.090]	−0.002 [−0.007, +0.003]
75%	+0.037 [−0.013, +0.086]	−0.003 [−0.008, +0.001]
100%	+0.039 [−0.010, +0.091]	−0.004 [−0.010, +0.002]

Retention drives the harm when it is incomplete At 50% overlap, dilating the mask from 0 to 25 px raises retention from 0.500 to 0.946 and moves the masking effect from -0.097 $[-0.164, -0.031]$ to -0.153 $[-0.221, -0.089]$. At 100% overlap the ruler is already retained (0.999) and the effect stays flat (-0.129 $[-0.195, -0.066]$ without and -0.138 $[-0.200, -0.082]$ with a 50 px margin), although dilation *lowers* the ruler’s share of visible pixels (0.055 to 0.018). Harm therefore tracks whether the ruler is visible, not how much of the image it occupies once visible (Table 5). The intervals of neighbouring margins overlap; the pattern, not each step, is the finding.

Table 5: Mask dilation: restoring context around the lesion restores ruler pixels (retention) and changes harm.

Overlap	Margin (px)	Ruler retained	Ruler share of visible pixels	mask – ERM (rev)
50%	0	0.500	0.028	-0.097 $[-0.164, -0.031]$
	10	0.725	0.029	-0.138 $[-0.203, -0.074]$
	25	0.946	0.026	-0.153 $[-0.221, -0.089]$
	50	1.000	0.018	-0.142 $[-0.206, -0.083]$
75%	0	0.750	0.042	-0.111 $[-0.181, -0.044]$
	10	0.940	0.037	-0.152 $[-0.217, -0.091]$
	25	1.000	0.028	-0.160 $[-0.226, -0.096]$
	50	1.000	0.018	-0.140 $[-0.201, -0.079]$
100%	0	0.999	0.055	-0.129 $[-0.195, -0.066]$
	10	1.000	0.040	-0.144 $[-0.206, -0.082]$
	25	1.000	0.028	-0.144 $[-0.209, -0.082]$
	50	1.000	0.018	-0.138 $[-0.200, -0.082]$

The masked model relies more on the retained artifact On the same images, inserting the ruler at full overlap changes the masked head’s prediction by 0.491 on average and the ERM head’s by 0.235 — about 2.1 times as much; the paired difference is $+0.255$ $[+0.221, +0.287]$ (Table 6, Figure 3). At 0% overlap the masked head cannot see the ruler (0.000) whereas ERM still reacts to it (0.128). This is the mechanism in one measurement: masking removes the competing evidence, the retained artifact becomes the most informative remaining feature, and the head leans on it. The counterfactual was saved for the dermoscopy sweep only; the other sweeps did not store per-image counterfactual scores, so this measurement is not replicated across modalities.

Table 6: Same-head counterfactual sensitivity to inserting the artifact (reversed test, artifact-bearing images), crossed CIs of the paired difference.

Sweep	Overlap	$ \Delta p $ ERM	$ \Delta p $ mask	mask – ERM
dermoscopy ruler (DINOv2)	0%	0.128	0.000	-0.128 $[-0.136, -0.121]$
	25%	0.190	0.353	$+0.162$ $[+0.136, +0.188]$
	50%	0.195	0.399	$+0.204$ $[+0.177, +0.230]$
	75%	0.197	0.422	$+0.225$ $[+0.195, +0.255]$
	100%	0.235	0.491	$+0.255$ $[+0.221, +0.287]$

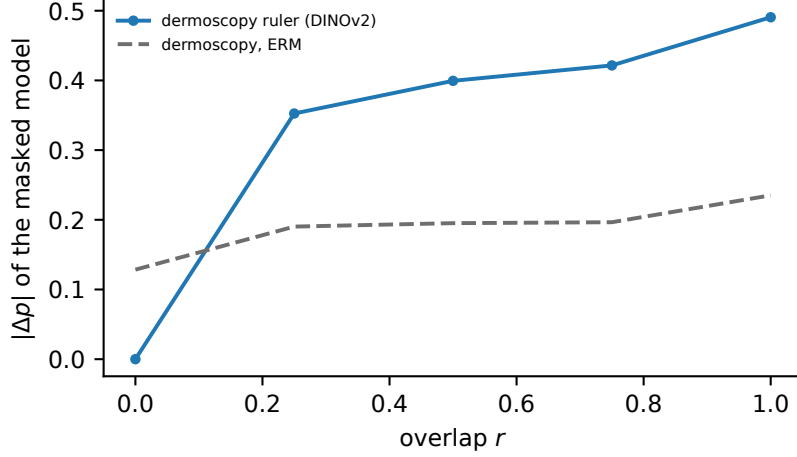


Figure 3: Mean same-head sensitivity $|\Delta p|$ to inserting the ruler, masked model (solid) and ERM (dashed), by overlap (dermoscopy sweep).

Lesion size Table 7 splits the test set by lesion area. The ruler covers a median 0.012 of a large lesion’s area, 0.031 of a medium and 0.106 of a small one. With the corrected intervals, the harm at full overlap is significant for small (-0.364 $[-0.509, -0.217]$) and medium lesions (-0.158 $[-0.295, -0.027]$) and not for large ones ($+0.063$ $[-0.015, +0.144]$, which does not differ significantly from zero); at 50% overlap it is significant only for small lesions (-0.326 $[-0.471, -0.176]$; medium -0.129 $[-0.283, +0.027]$). What survives is an ordering — harm is largest where the artifact is large relative to the lesion — with wide intervals because each tertile has few melanomas. No fixed ratio threshold is claimed.

Table 7: Lesion-size strata of the controlled sweep (reversed-test AUROC, crossed CIs).

Lesion tertile	Ruler/lesion area (median)	Melanomas in test	mask – ERM at 0%	50%	100%
large	0.012	36	+0.249 [+0.155, +0.346]	+0.087 [+0.006, +0.172]	+0.063 [−0.015, +0.144]
medium	0.031	21	+0.215 [+0.033, +0.391]	−0.129 [−0.283, +0.027]	−0.158 [−0.295, −0.027]
small	0.106	16	+0.150 [+0.036, +0.273]	−0.326 [−0.471, −0.176]	−0.364 [−0.509, −0.217]

5.3 Other modalities

Table 8 repeats the sweep with synthetic artifacts on real images from four further modalities. Across all 11 controlled sweeps of this report (four dermoscopy, seven others), the location interaction is positive with an interval above zero in 10 and negative in 1. Whether the falling benefit turns into harm at full overlap depends on the setting: masking harms significantly at full overlap in 7 sweeps, still helps in 2 and makes no significant difference in 2.

Table 8: Controlled sweeps in the other modalities (synthetic caliper, debris, tube pasted at controlled overlap; reversed-test AUROC, crossed CIs).

Sweep	mask – ERM at 0%	at 100%	Location interaction
Thyroid caliper, DINOv2	+0.402 [+0.315, +0.489]	−0.030 [−0.105, +0.042]	+0.433 [+0.362, +0.507]
Thyroid caliper, MedSigLIP	+0.514 [+0.430, +0.600]	+0.044 [−0.011, +0.104]	+0.470 [+0.388, +0.556]
Ovarian caliper, DINOv2	+0.041 [−0.096, +0.175]	−0.197 [−0.357, −0.035]	+0.238 [+0.144, +0.340]
Capsule debris, DINOv2	+0.116 [+0.052, +0.191]	−0.112 [−0.209, −0.019]	+0.228 [+0.144, +0.324]
Capsule debris, MedSigLIP	+0.374 [+0.248, +0.509]	+0.216 [+0.118, +0.317]	+0.158 [+0.012, +0.314]
Chest tube, RAD-DINO	−0.002 [−0.010, +0.005]	+0.111 [+0.095, +0.127]	−0.113 [−0.129, −0.097]
Chest tube, DINOv2	+0.254 [+0.230, +0.278]	−0.083 [−0.112, −0.057]	+0.337 [+0.312, +0.364]

- **Thyroid calipers.** Masking helps strongly with the caliper outside the nodule (DINOv2 +0.402

[+0.315, +0.489]; MedSigLIP +0.514 [+0.430, +0.600]) and the benefit vanishes at full overlap (−0.030 [−0.105, +0.042]; +0.044 [−0.011, +0.104]): the law holds as a loss of benefit, not as harm.

- **Ovarian calipers.** Masking harms at full overlap (−0.197 [−0.357, −0.035]) and its benefit outside the tumour is not significant (+0.041 [−0.096, +0.175]): MMOTU tumours fill much of the frame, so little background is removed.
- **Capsule debris.** With DINOv2 the sweep reverses (+0.116 [+0.052, +0.191] to −0.112 [−0.209, −0.019]). With MedSigLIP masking still helps at full overlap (+0.216 [+0.118, +0.317]) but less than outside (+0.374 [+0.248, +0.509]; interaction +0.158 [+0.012, +0.314]): the box ROI contains much normal mucosa, so masking removes little of the lesion’s context.
- **Chest tube — the boundary.** With RAD-DINO the ERM model ignores a tube outside the lungs (reversed 0.913 versus clean 0.915 at no overlap), so masking has nothing to remove at $r = 0$; at full overlap ERM collapses (0.614) and the masked model less so (0.725). The interaction is negative. With DINOv2, which does use the extra-pulmonary tube (reversed 0.629 versus clean 0.859), the law holds (+0.337 [+0.312, +0.364]). The law’s premise is that the encoder uses the out-of-ROI artifact; when it does not, masking cannot help outside and the law does not apply.

6 What failed or changed

- **Two lesion-size cells lost significance with the corrected bootstrap:** large lesions at full overlap and medium lesions at 50% overlap. Earlier drafts described harm as increasing monotonically across all three tertiles at both overlaps; that statement is withdrawn and replaced by the ordering above. The old and new intervals are listed in `results/bootstrap_correction/sweeps_umte_old_vs_new.md`.
- **Harm at full overlap is not universal.** Earlier drafts said masking “harms once the artifact overlaps the ROI”. With corrected intervals, the robust statement is the *interaction* (10 of 11); harm at full overlap is significant in 7 of 11 sweeps.
- **Chest radiography with RAD-DINO contradicts the law**, as a failure of its premise (above). It was registered as a confirmation and is reported as a boundary case.
- **The counterfactual was not saved for the non-dermoscopy sweeps**, so the mechanism measurement is dermoscopy-only.
- **The ovarian sweep has no significant benefit outside the ROI**, so its crossover rests on the harm side only.

7 Conclusion

Supported. The location law in its interaction form (Q2a) holds in 10 of 11 controlled sweeps across five modalities and four encoders. The mechanism (Q2b) is retention: occlusion alone does not harm, harm follows the visibility of the artifact under dilation, and the masked head’s same-image reliance on an in-ROI artifact is about 2.1 times that of ERM (+0.255 [+0.221, +0.287]).

Qualified. Harm at full overlap is setting-dependent; the lesion-size gradient is an ordering with two non-significant cells. **Not supported.** The law for an encoder that ignores the out-of-ROI artifact (chest tube, RAD-DINO). The practical reading is: masking removes what lies outside the ROI and concentrates the model on what lies inside; whether that helps depends on where the shortcut sits.

8 Questions for the supervisor

1. Should the thesis state the law in its interaction form only (robust) and treat harm at full overlap as setting-dependent, or keep harm as the headline for dermoscopy, where it holds with every encoder?
2. Is one modality (dermoscopy) enough for the counterfactual mechanism, or should the counterfactual be re-run for thyroid and capsule before submission (about one GPU-hour each)?
3. Is it acceptable to present chest radiography with RAD-DINO as a boundary of the law rather than as a failed confirmation, given that it was registered as a confirmation?

9 Next stage

Stage 3 leaves synthetic artifacts. It asks whether the same crossover appears with *real* artifacts — hair in dermoscopy, calipers in thyroid and ovarian ultrasound, debris in capsule endoscopy, devices in chest radiographs — where location is defined by real masks and detectors. It also reports the overlap-contrast design that failed (E12), the detector validity checks and the 16-test confirmatory family with Holm correction.

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